Cognizant-Up Grad Training Program UGE_DOTNET FSD (Python)

Weekly Submission Document Trainee Information

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Batch Name: B1 Python

Course: UGE DOTNet FSD (Python)

Training Duration: 16 Feb 2026-26" May 2026 Session

Timings: 9:30 AM-6:30 PM

Faculty: Kunal

Week-7

Q1: Notification System (Strategy + DI)

✓ Question (Short)

Design a system to send Email/SMS/Push notifications

- No `if-else`
- Add new types easily
- Retry once if fail

```
// Step 1: Common Interface
public interface INotification
{
    Task SendAsync(string message);
}

// Step 2: Implementations
public class EmailNotification : INotification
{
    public async Task SendAsync(string message)
    {
        Console.WriteLine("Email Sent: " + message);
```

```
        await Task.CompletedTask;
    }
}

public class SmsNotification : INotification
{
    public async Task SendAsync(string message)
    {
        Console.WriteLine("SMS Sent: " + message);
        await Task.CompletedTask;
    }
}

// Step 3: Factory (No if-else → use Dictionary)
public class NotificationFactory
{
    private readonly Dictionary<string, INotification> _services;

    public NotificationFactory(IEnumerable<INotification> services)
    {
        _services = services.ToDictionary(s => s.GetType().Name);
    }

    public INotification Get(string type)
    {
        if (_services.ContainsKey(type))
            return _services[type];

        throw new Exception("Invalid Type");
    }
}

// Step 4: Service with Retry
public class NotificationService
{
    private readonly NotificationFactory _factory;

    public NotificationService(NotificationFactory factory)
    {
        _factory = factory;
    }

    public async Task Send(string type, string msg)
    {
        var service = _factory.Get(type);

        try
        {
```

```
            await service.SendAsync(msg);
        }
        catch
        {
            Console.WriteLine("Retrying...");
            await service.SendAsync(msg);
        }
    }
}
```

Q2: Thread-Safe Cache

✓ Question (Short)

Design cache:

- Set(key, value, expiry)
- Thread-safe
- Auto remove expired

```
public class CacheItem
{
    public string Value { get; set; }
    public DateTime Expiry { get; set; }
}

public class CacheService
{
    private static readonly object _lock = new();
    private static Dictionary<string, CacheItem> _cache = new();

    public void Set(string key, string value, int seconds)
    {
        lock (_lock)
        {
            _cache[key] = new CacheItem
            {
                Value = value,
                Expiry = DateTime.Now.AddSeconds(seconds)
            };
        }
    }
```

```
    public string Get(string key)
    {
        lock (_lock)
        {
            if (_cache.ContainsKey(key))
            {
                var item = _cache[key];

                if (item.Expiry > DateTime.Now)
                    return item.Value;

                _cache.Remove(key);
            }
            return null;
        }
    }
}
```

Q3: JWT Authentication (Basic)

✓ Question (Short)

Login → generate JWT
Protected API → validate token

```
// Generate Token
public string GenerateToken(string userId)
{
    var key = new SymmetricSecurityKey(Encoding.UTF8.GetBytes("secretkey123"));
    var creds = new SigningCredentials(key, SecurityAlgorithms.HmacSha256);

    var token = new JwtSecurityToken(
        claims: new[] { new Claim("UserId", userId) },
        expires: DateTime.Now.AddMinutes(15),
        signingCredentials: creds);

    return new JwtSecurityTokenHandler().WriteToken(token);
}
```

Q4: Rate Limiter

✓ Question (Short)

Limit 5 requests per 10 sec

```
public class RateLimiter
{
    private Dictionary<string, List<DateTime>> _requests = new();

    public bool IsAllowed(string user)
    {
        if (!_requests.ContainsKey(user))
            _requests[user] = new List<DateTime>();

        var now = DateTime.Now;

        _requests[user].RemoveAll(t => (now - t).TotalSeconds > 10);

        if (_requests[user].Count >= 5)
            return false;

        _requests[user].Add(now);
        return true;
    }
}
```

Q5: Deadlock Example

✓ Code (Deadlock)

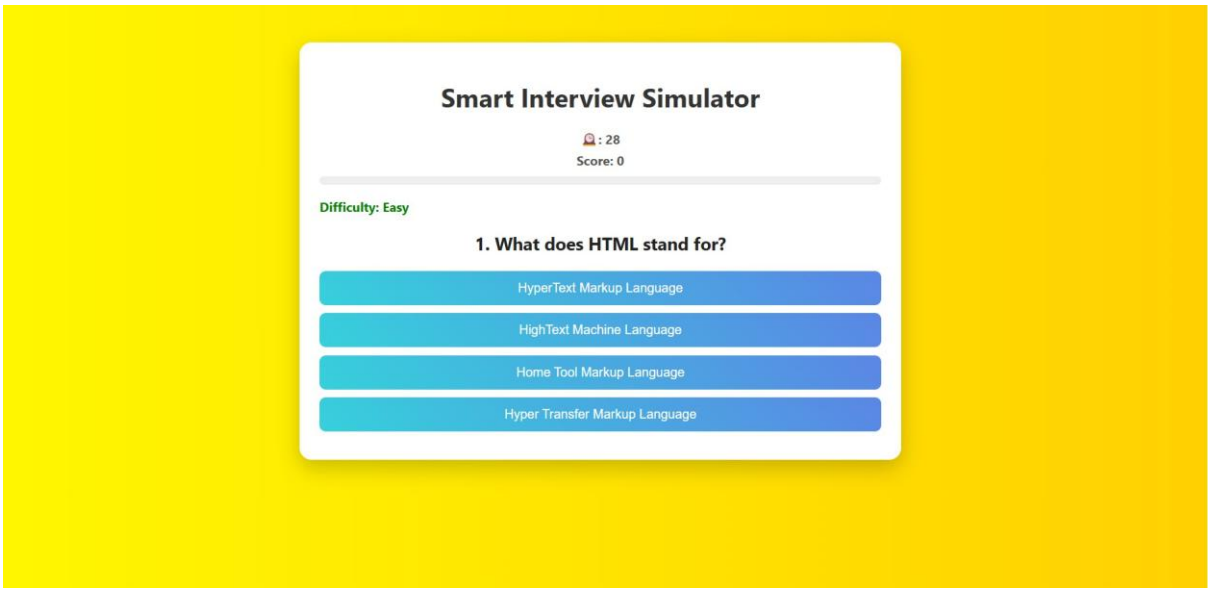
```
object lock1 = new object();
object lock2 = new object();

void Method1()
{
    lock (lock1)
    {
        Thread.Sleep(100);
        lock (lock2)
        {
            Console.WriteLine("Method1");
        }
    }
}

void Method2()
{
    lock (lock2)
    {
        Thread.Sleep(100);
        lock (lock1)
```

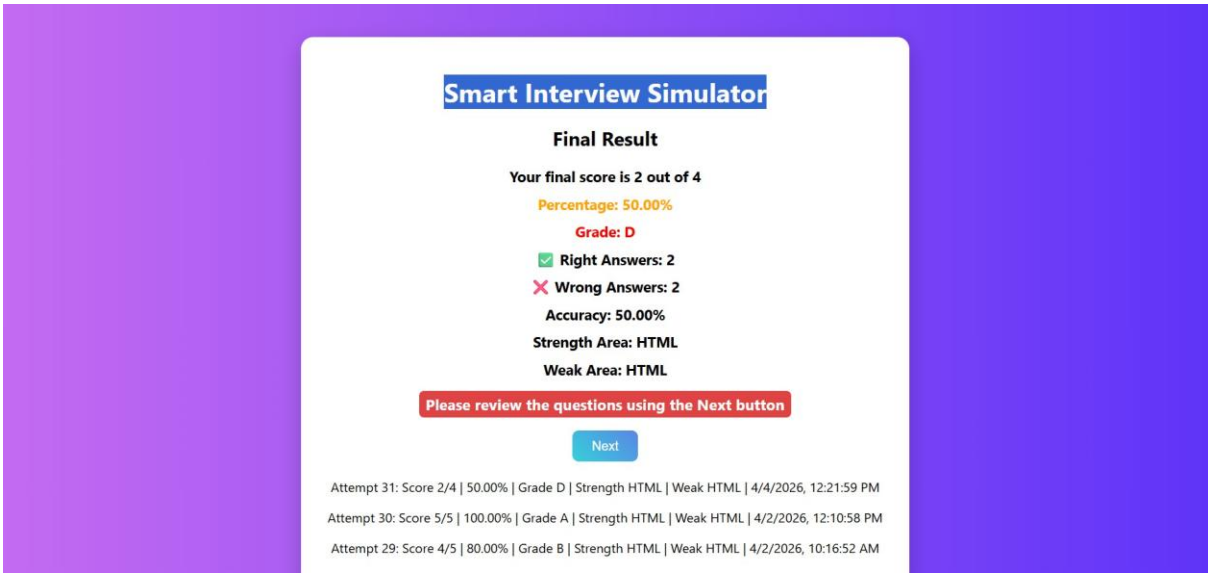
```
    {
        Console.WriteLine("Method2");
    }
  }
}
```

Q6: Custom HashMap (Simple)

```
public class MyHashMap
{
   private List<(int key, string value)>[] buckets;

   public MyHashMap(int size)
   {
      buckets = new List<(int, string)>[size];
   }

   private int Hash(int key) => key % buckets.Length;

   public void Add(int key, string value)
   {
      int index = Hash(key);

      if (buckets[index] == null)
         buckets[index] = new List<(int, string)>();

      buckets[index].Add((key, value));
   }

   public string Get(int key)
   {
      int index = Hash(key);

      if (buckets[index] != null)
      {
         foreach (var item in buckets[index])
         {
            if (item.key == key)
               return item.value;
         }
      }
      return null;
   }
}
```

Smart Interview Simulator





Full-Stack Healthcare Appointment System (ASP.NET Core + SQL)

🧩 Problem Statement

An emerging healthcare startup wants to transform its basic appointment booking website into a full-stack enterprise-grade application using ASP.NET Core 8, C#, and SQL Server.

Develop a multi-layered web application that supports doctor management, appointment scheduling, patient records, and secure data storage using MVC architecture and REST APIs.

The system must:

- Persist data in a relational database
- Use Entity Framework Core for ORM
- Follow SOLID principles and design patterns
- Implement authentication-ready structure and secure coding practices
- Support scalable and maintainable architecture

🧱 Core Features to Implement

Application Modules

- Dashboard Module – Displays upcoming appointments and statistics
- Doctors Module – CRUD operations for doctors and specialization
- Appointments Module – Book, update, cancel appointments
- Patient Profile Module – Stores patient details and history
- API Module – REST APIs for doctors, appointments, and patients

---

⚙ Functional Requirements

🗄 Database & SQL (Detailed Schema Required)

📌 You MUST create the following tables:

1. Patients

```
PatientId (PK, INT, Identity)
FullName (VARCHAR)
Email (UNIQUE)
PhoneNumber
CreatedAt (DATETIME)
```

---

2. Doctors

```
DoctorId (PK)
FullName
Specialization
Email
```

---

3. Appointments

```
AppointmentId (PK)
PatientId (FK → Patients.PatientId)
DoctorId (FK → Doctors.DoctorId)
AppointmentDate (DATETIME)
Status (Scheduled / Completed / Cancelled)
```

---

4. MedicalRecords

```
RecordId (PK)
PatientId (FK)
DoctorId (FK)
Diagnosis
Prescription
CreatedAt
```

---

5. Departments

```
DepartmentId (PK)
Name
```

---

6. DoctorDepartments

```
Id (PK)
DoctorId (FK)
DepartmentId (FK)
```

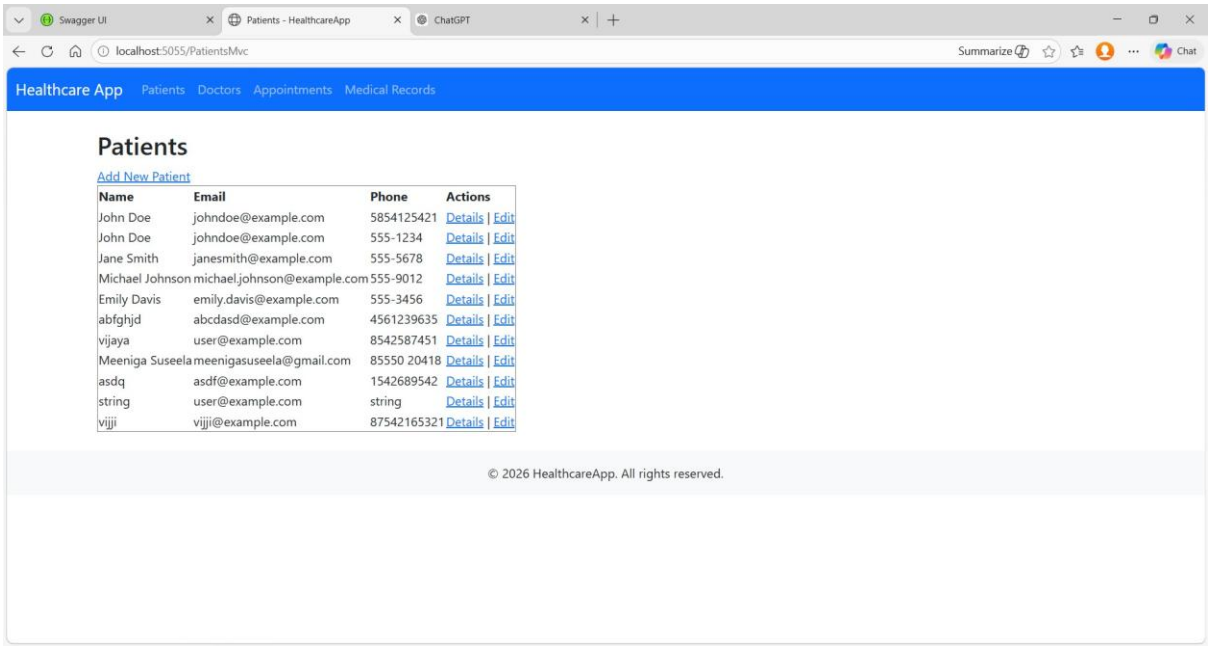
🔗 Relationships (MANDATORY)

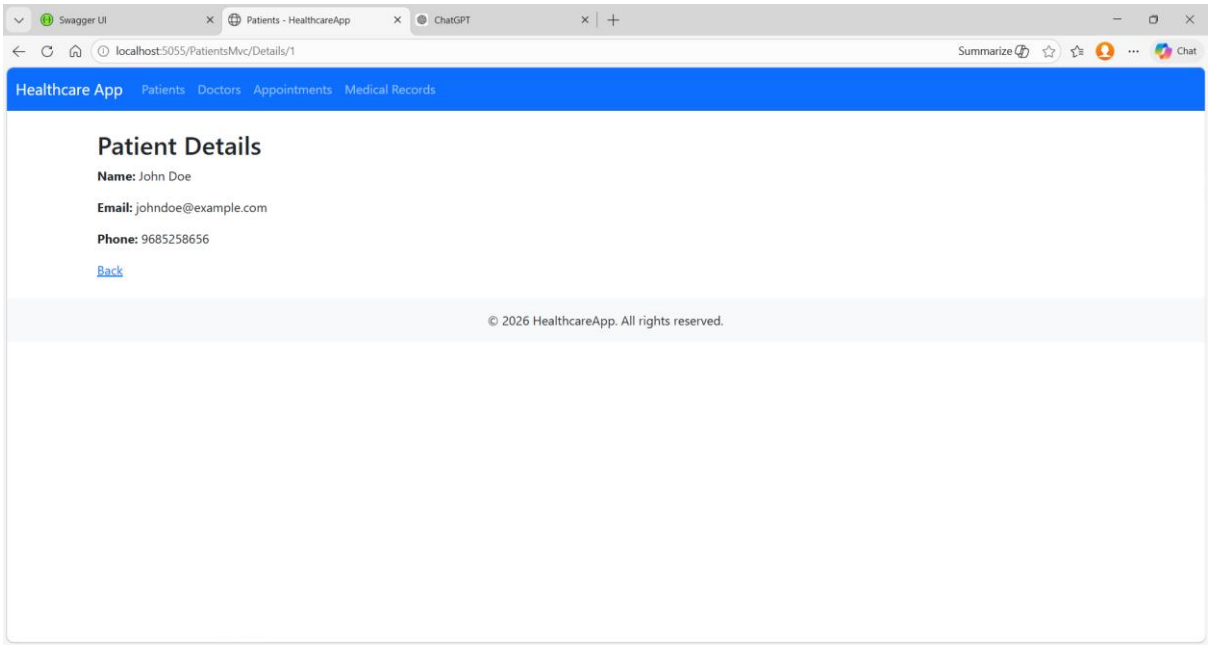
- One Patient → Many Appointments
- One Doctor → Many Appointments
- One Patient → Many Medical Records
- Many Doctors ↔ Many Departments

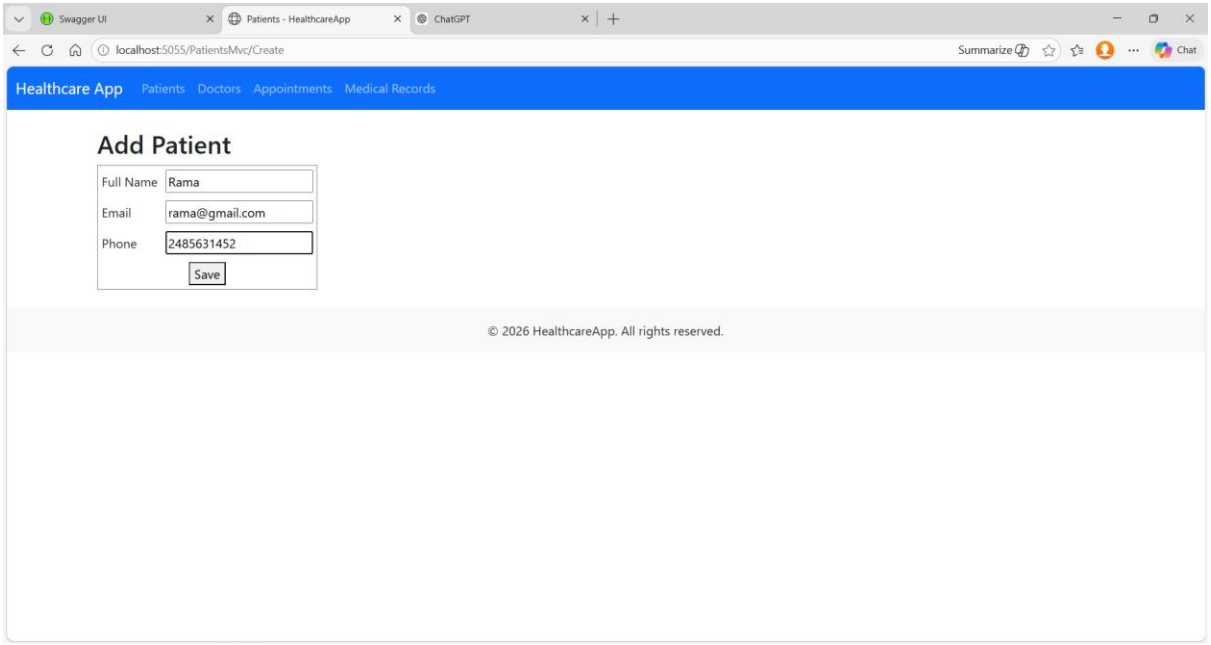
📌 SQL Requirements

You MUST write queries for:

- SELECT, WHERE, ORDER BY
- INNER JOIN, LEFT JOIN
- GROUP BY (appointments per doctor)
- Subqueries (patients with highest visits)
- UNION (combine doctor & patient emails)
- INSERT, UPDATE, DELETE






Swagger UI
Patients - HealthcareApp
ChatGPT
localhost:5055/PatientsMvc/Create
Summarize
Chat
Healthcare App
Patients
Doctors
Appointments
Medical Records
Add Patient
Full Name
Rama
Email
rama@gmail.com
Phone
2485631452
Save
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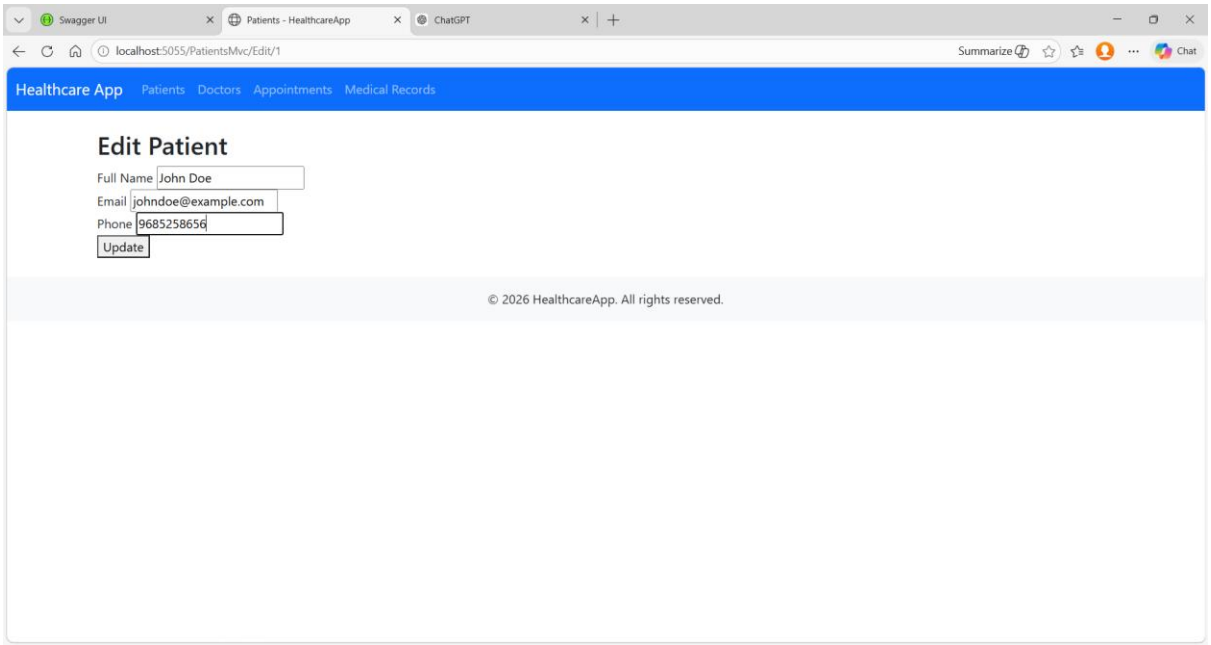
Swagger UI
Patients - HealthcareApp
ChatGPT
localhost:5055/PatientsMvc/Edit/1
Summarize
Chat
Healthcare App
Patients
Doctors
Appointments
Medical Records
Edit Patient
Full Name
John Doe
Email
johndoe@example.com
Phone
9685258656
Update
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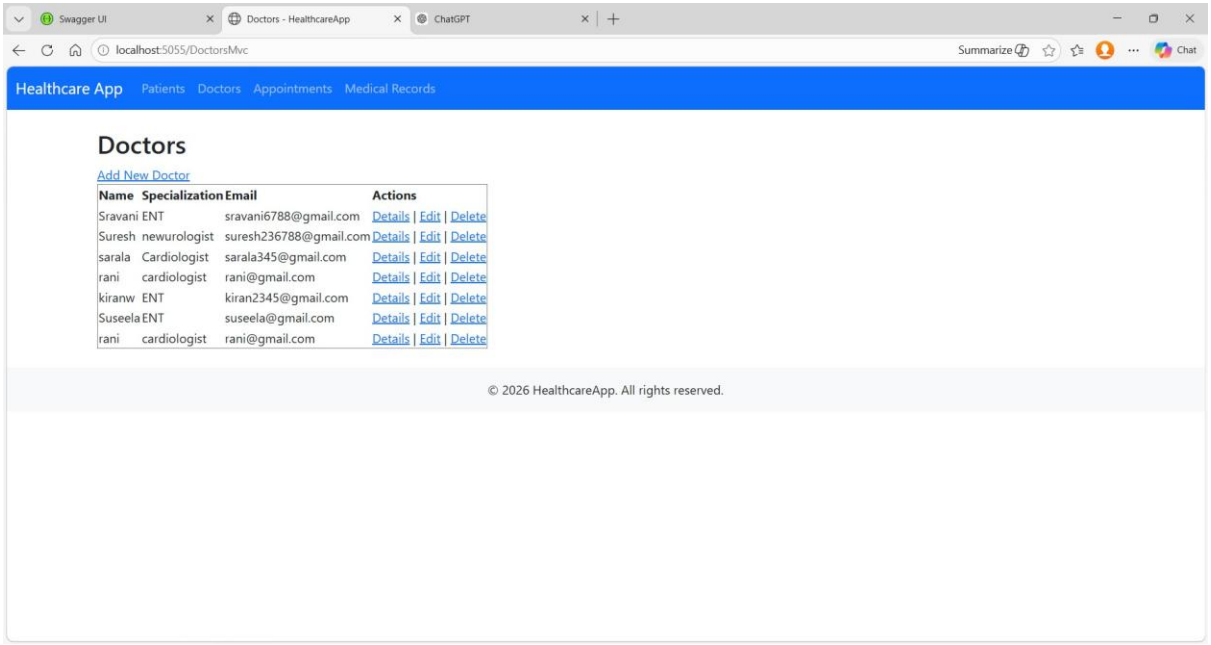
Swagger UI
Doctors - HealthcareApp
ChatGPT
localhost:5055/DoctorsMvc
Summarize
Chat
Healthcare App
Patients
Doctors
Appointments
Medical Records
Doctors
Add New Doctor
Name Specialization Email Actions
Sravani ENT sravani6788@gmail.com Details | Edit | Delete
Suresh newurologist suresh236788@gmail.com Details | Edit | Delete
sarala Cardiologist sarala345@gmail.com Details | Edit | Delete
rani cardiologist rani@gmail.com Details | Edit | Delete
kiranw ENT kiran2345@gmail.com Details | Edit | Delete
Suseela ENT suseela@gmail.com Details | Edit | Delete
rani cardiologist rani@gmail.com Details | Edit | Delete
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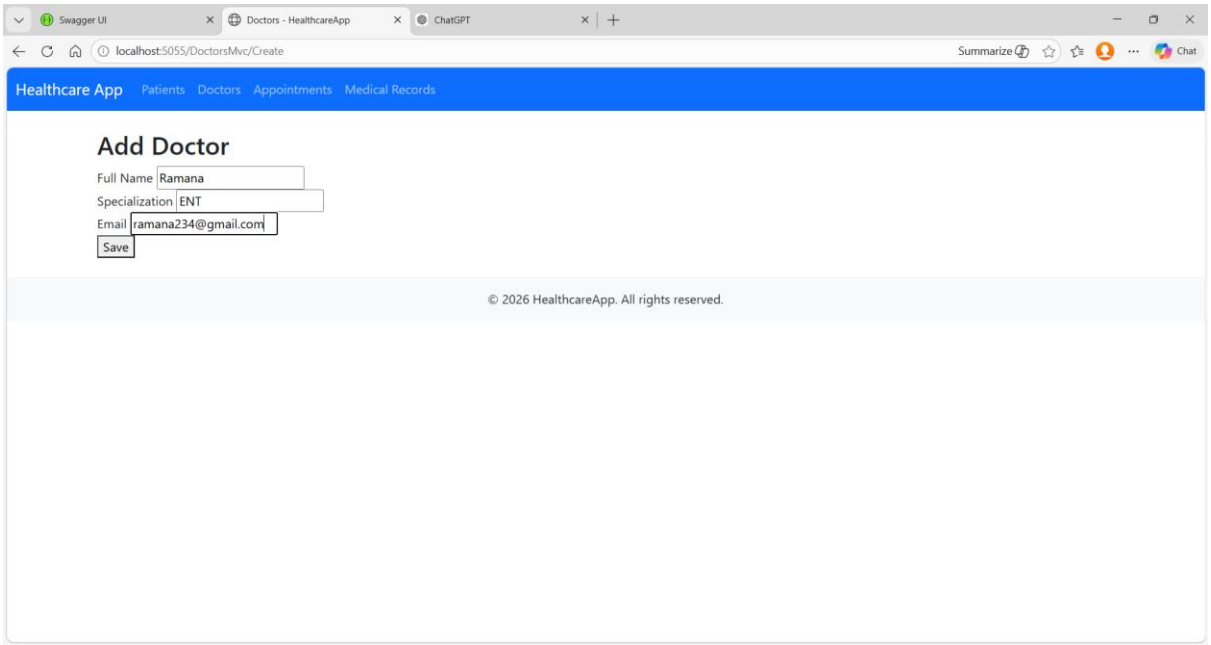
Swagger UI
Doctors - HealthcareApp
ChatGPT
localhost:5055/DoctorsMvc/Create
Summarize
Chat
Healthcare App
Patients
Doctors
Appointments
Medical Records
Add Doctor
Full Name Ramana
Specialization ENT
Email ramana234@gmail.com
Save
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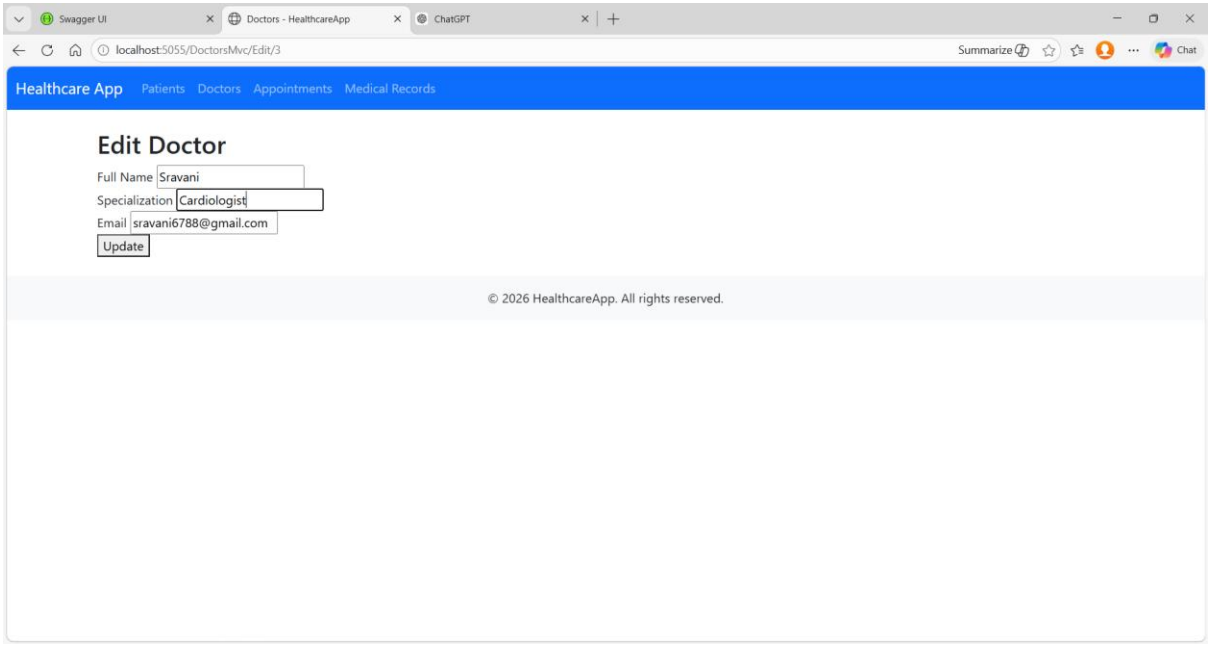
Swagger UI
Doctors - HealthcareApp
ChatGPT
localhost:5055/DoctorsMvc/Edit/3
Summarize
Chat
Healthcare App
Patients
Doctors
Appointments
Medical Records
Edit Doctor
Full Name Sravani
Specialization Cardiologist
Email sravani6788@gmail.com
Update
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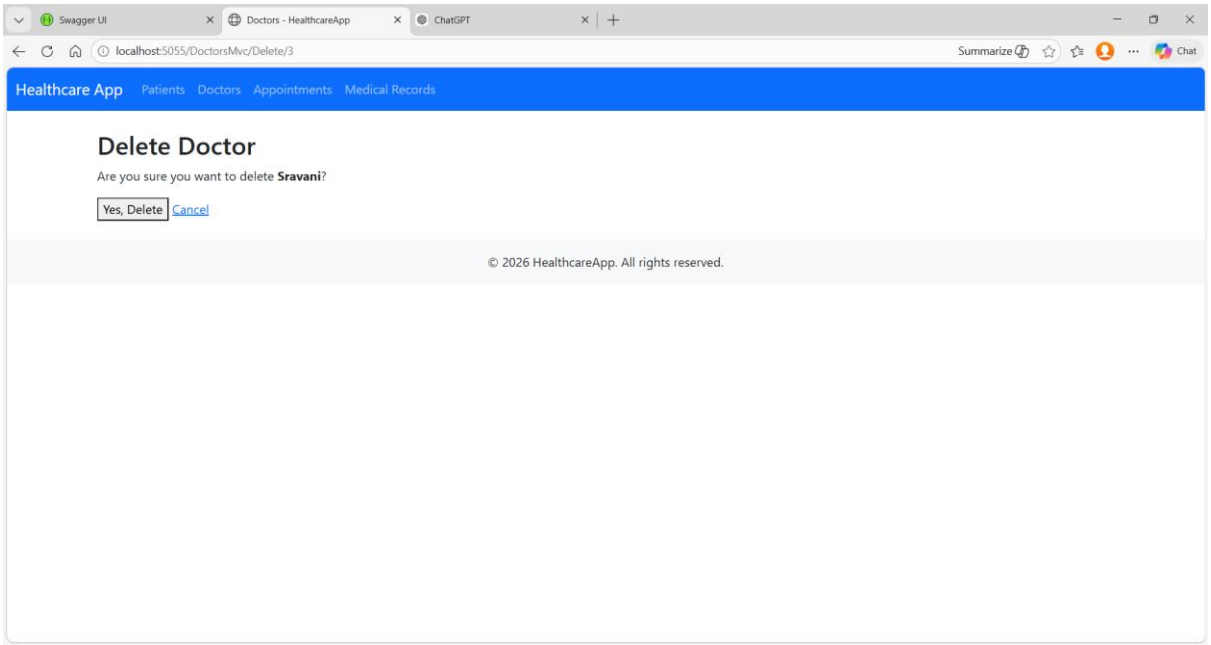
Swagger UI
Doctors - HealthcareApp
ChatGPT
localhost:5055/DoctorsMvc/Delete/3
Summarize
Chat
Healthcare App
Patients
Doctors
Appointments
Medical Records
Delete Doctor
Are you sure you want to delete Sravani?
Yes, Delete
Cancel
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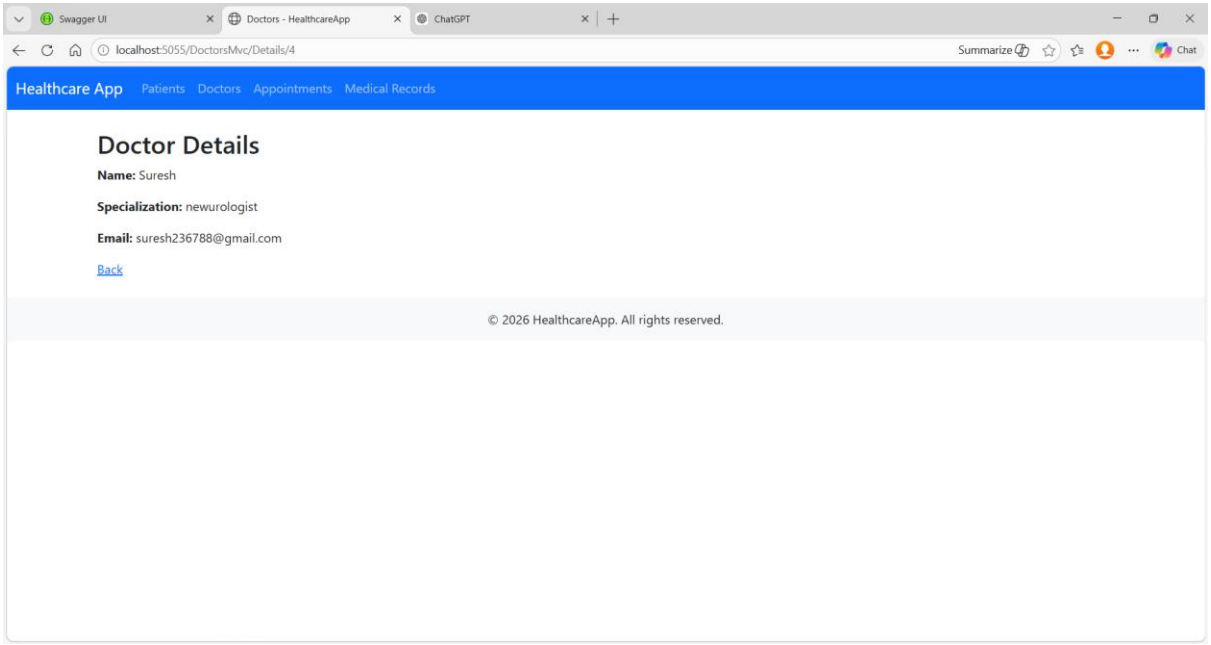
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ChatGPT
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Summarize
Chat
Healthcare App
Patients
Doctors
Appointments
Medical Records
Doctor Details
Name: Suresh
Specialization: newurologist
Email: suresh236788@gmail.com
Back
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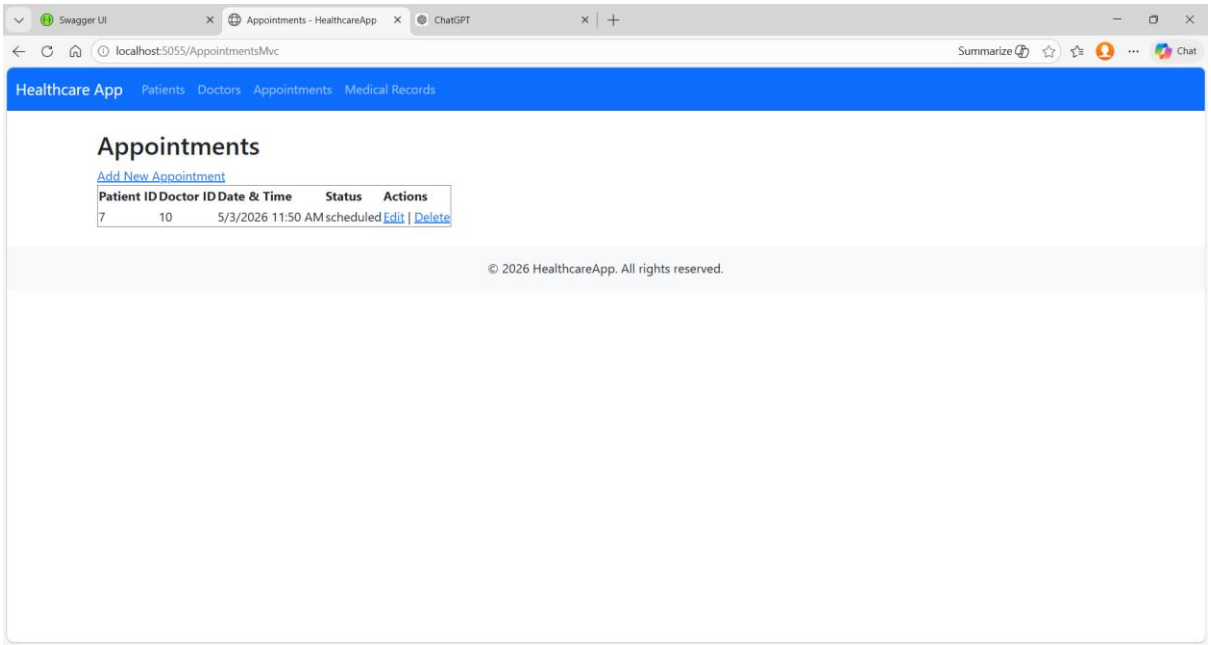
Swagger UI
Appointments - HealthcareApp
ChatGPT
localhost:5055/AppointmentsMvc
Summarize
Chat
Healthcare App
Patients
Doctors
Appointments
Medical Records
Appointments
Add New Appointment
Patient ID
Doctor ID
Date & Time
Status
Actions
7
10
5/3/2026 11:50 AM
scheduled
Edit | Delete
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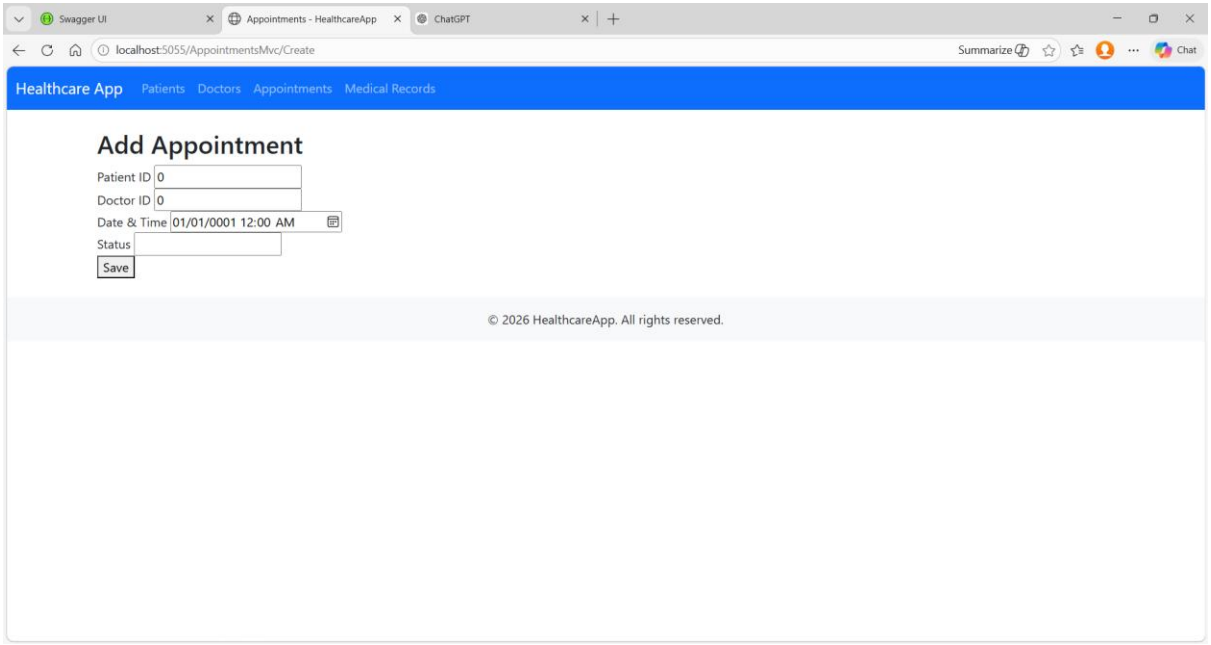
Swagger UI
Appointments - HealthcareApp
ChatGPT
localhost:5055/AppointmentsMvc/Create
Summarize
Chat
Healthcare App
Patients
Doctors
Appointments
Medical Records
Add Appointment
Patient ID 0
Doctor ID 0
Date & Time 01/01/0001 12:00 AM
Status
Save
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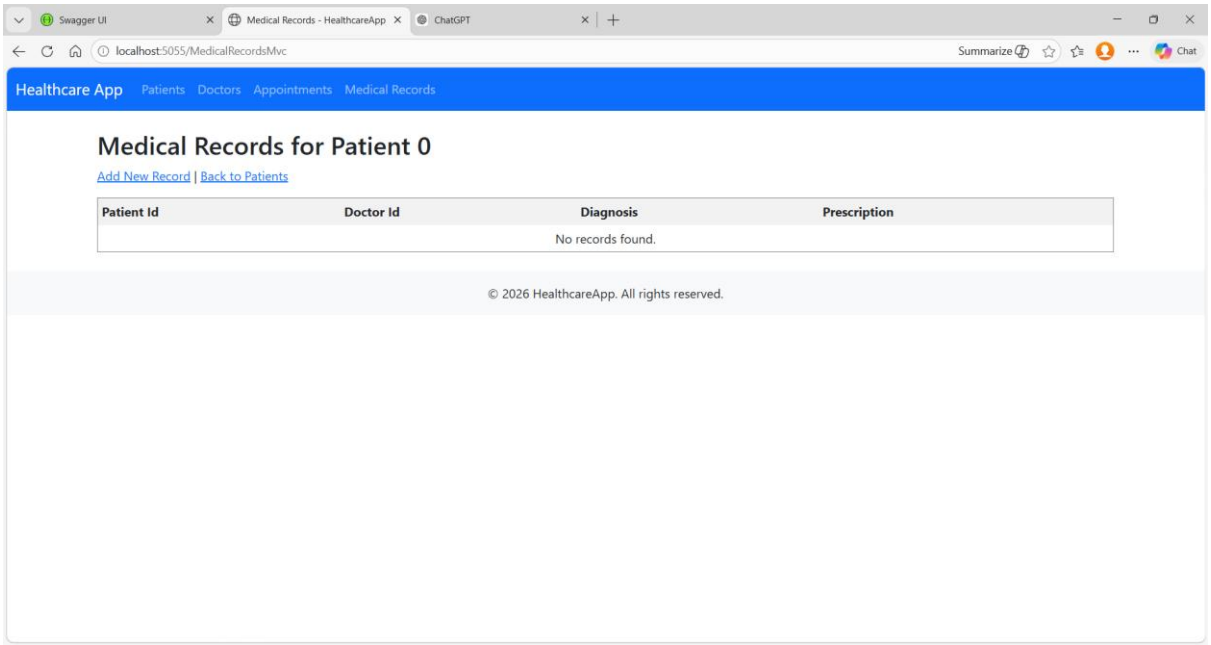
Swagger UI
Medical Records - HealthcareApp
ChatGPT
localhost:5055/MedicalRecordsMvc
Summarize
Chat
Healthcare App
Patients
Doctors
Appointments
Medical Records
Medical Records for Patient 0
Add New Record | Back to Patients
Patient Id
Doctor Id
Diagnosis
Prescription
No records found.
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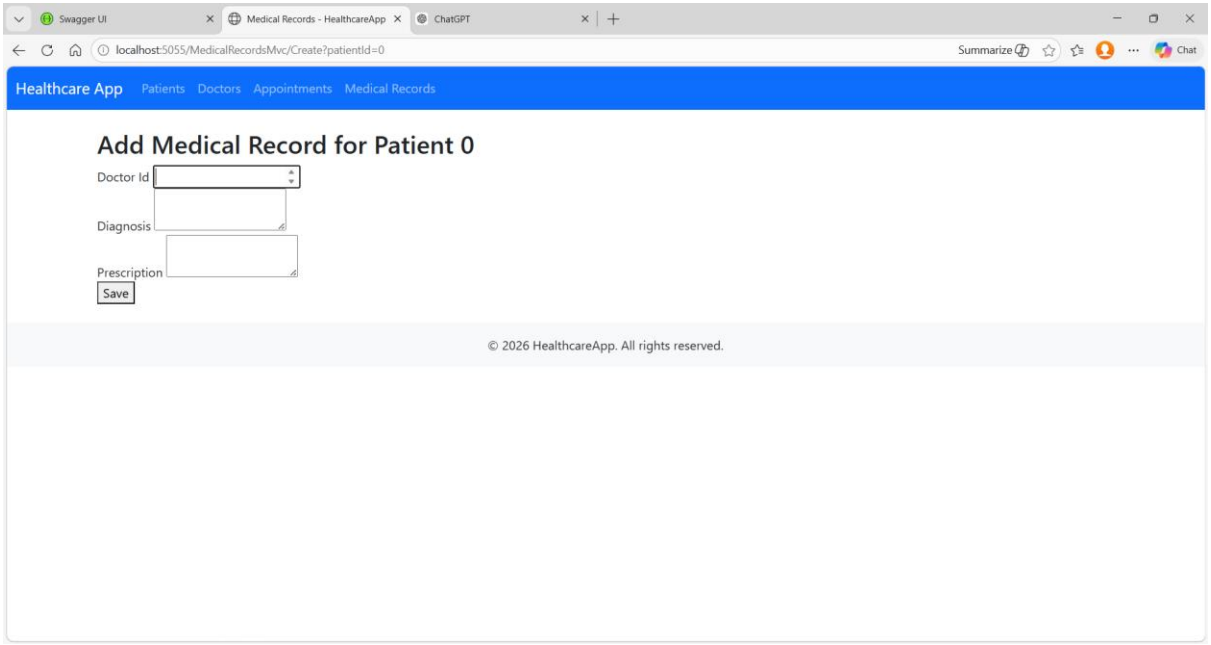
Swagger UI
Medical Records - HealthcareApp
ChatGPT
localhost:5055/MedicalRecordsMvc/Create?patientId=0
Summarize
Chat
Healthcare App
Patients
Doctors
Appointments
Medical Records
Add Medical Record for Patient 0
Doctor Id
Diagnosis
Prescription
Save
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